Exp :-3

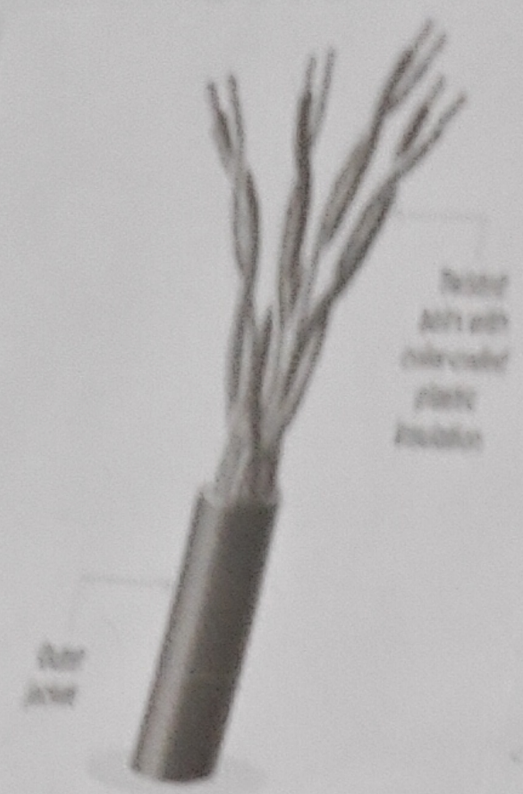
Study of different types of network cables

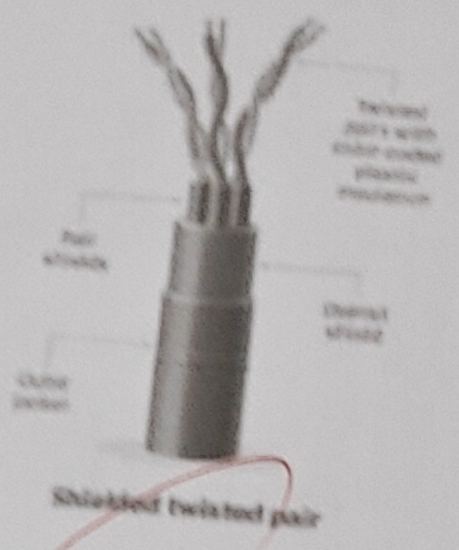
Aim :-
To study the different types of network cables.

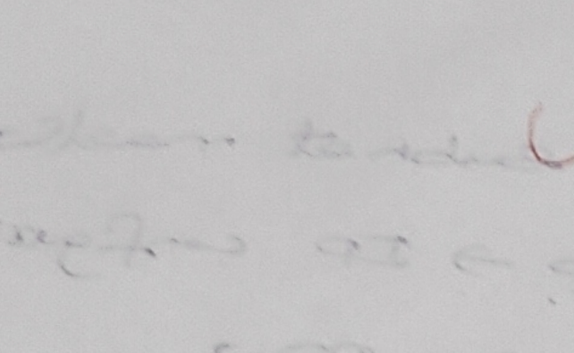
a. Understand different types of network cable :

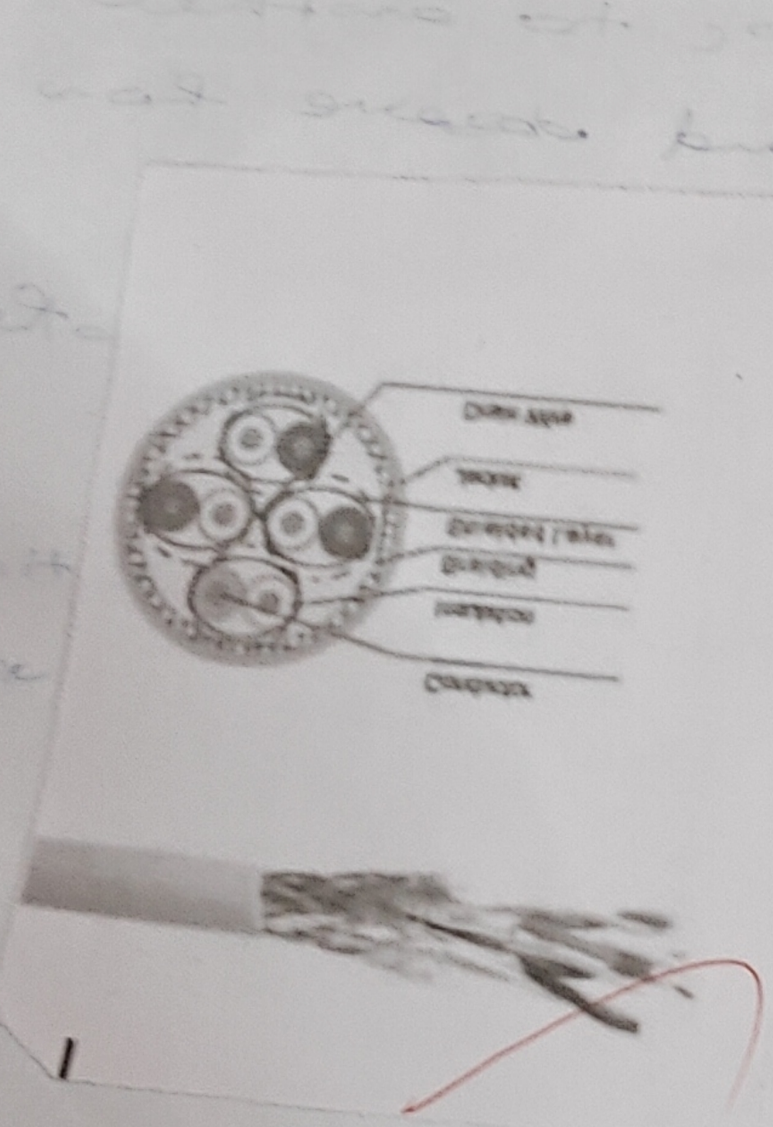
1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial cable
4. Fibre Optic cable

cable type	category	Maximum Data Transmission	Advantages/ Disadvantages	Application/ Use
UTP	category 3	10 Bps	Adv • cheaper in cost • Easy to install Disadv • more prone to EMI	10Base-T ethernet Fast ethernet, Gigabit ethernet
	category 5	Up to 100 Mbps		
	category 5e	1 Gbps		
STP	Category 6a	10 Gbps	Adv • shielded • Faster • less susceptible Disadv • Expensive • Greater installation	Gigabit ethernet 10G ethernet used in data centers Gigabit ethernet
SSTP	Categ-ory 7	10 Gbps		
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Adv • High bandwidth • versatile Disadv • cost • size bulky	speed of signal is soon Television network




Shielded twisted pair





Fibre optics cable	Single mode multi mode	100Gbps	Adv • High speed • High Bandwidth Disadv • Expensive • Skilled instructors	Maximum distance of fibre optics cable is around 100 metres

(b) Make your own ethernet cross-over cable/ straight cable :-

Tools and parts needed :

- Ethernet cabling - cat5e is certified for gigabit support
- A crimping tool - This is all in one networking tool and cut shielding off cables
- optional two plug shields

Step 1 :-

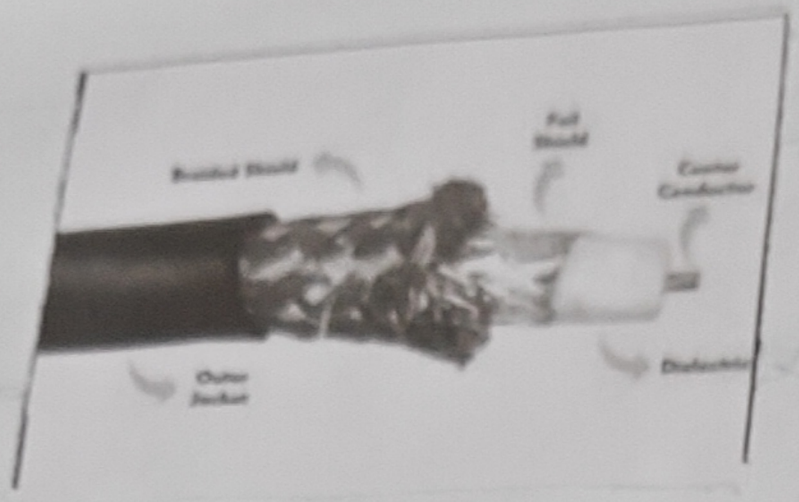
To start construction of device, Begin by threading shields onto cable.

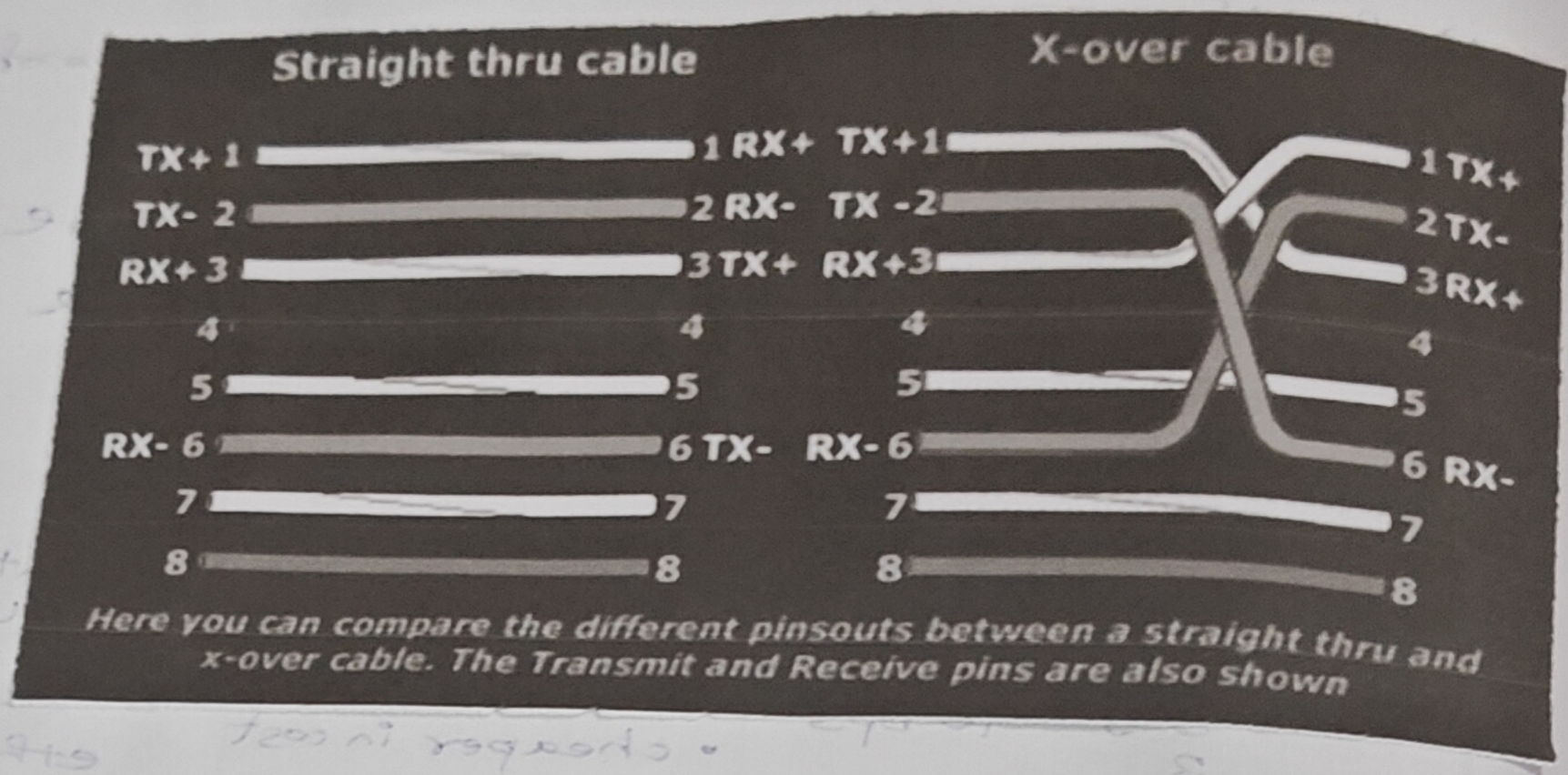
Step 2 :-

Next, strip approx. 1.5 cm of cable shielding from Both ends.

Step 3 :-

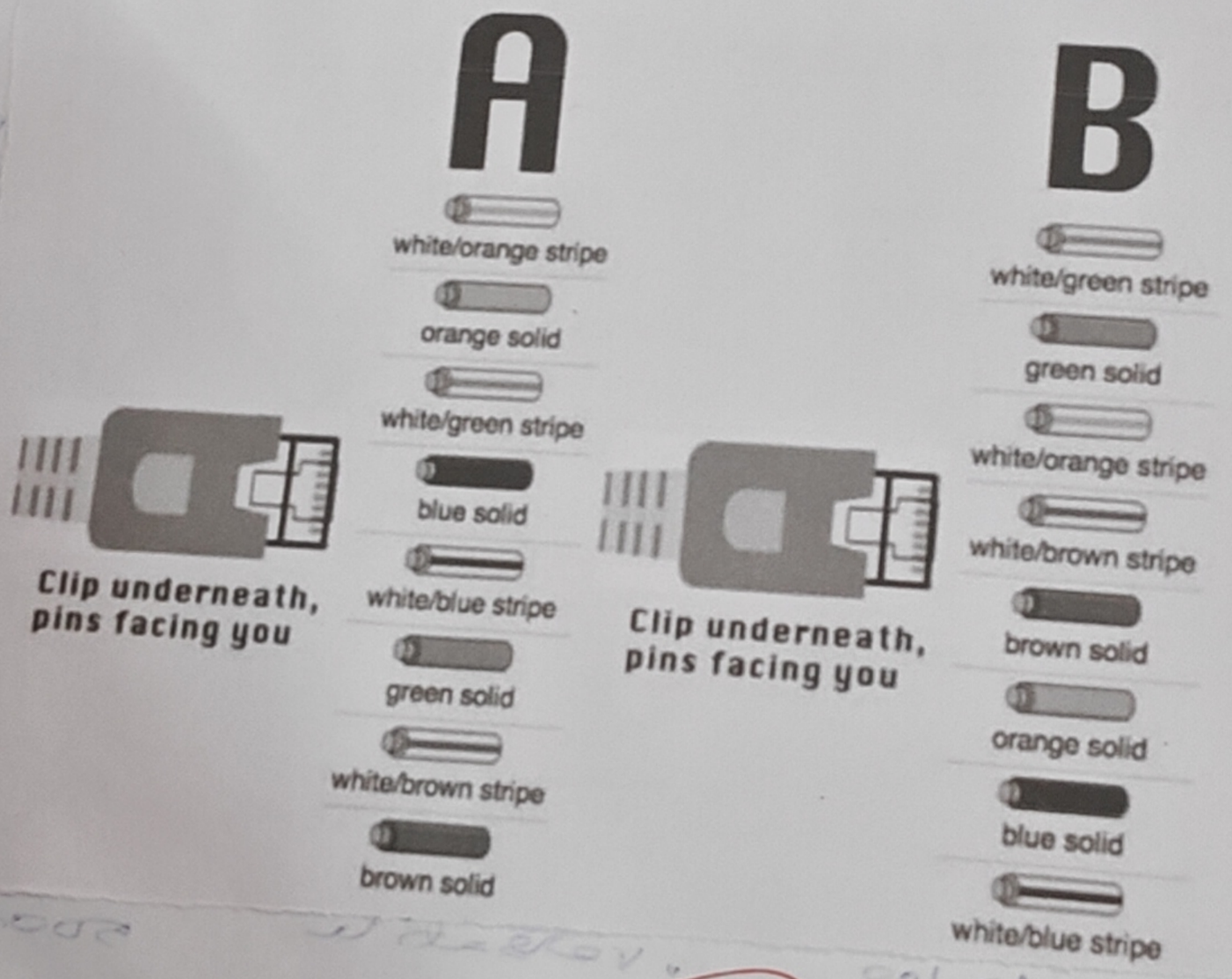
After, you will need to untangle wires; there should be four twisted pairs

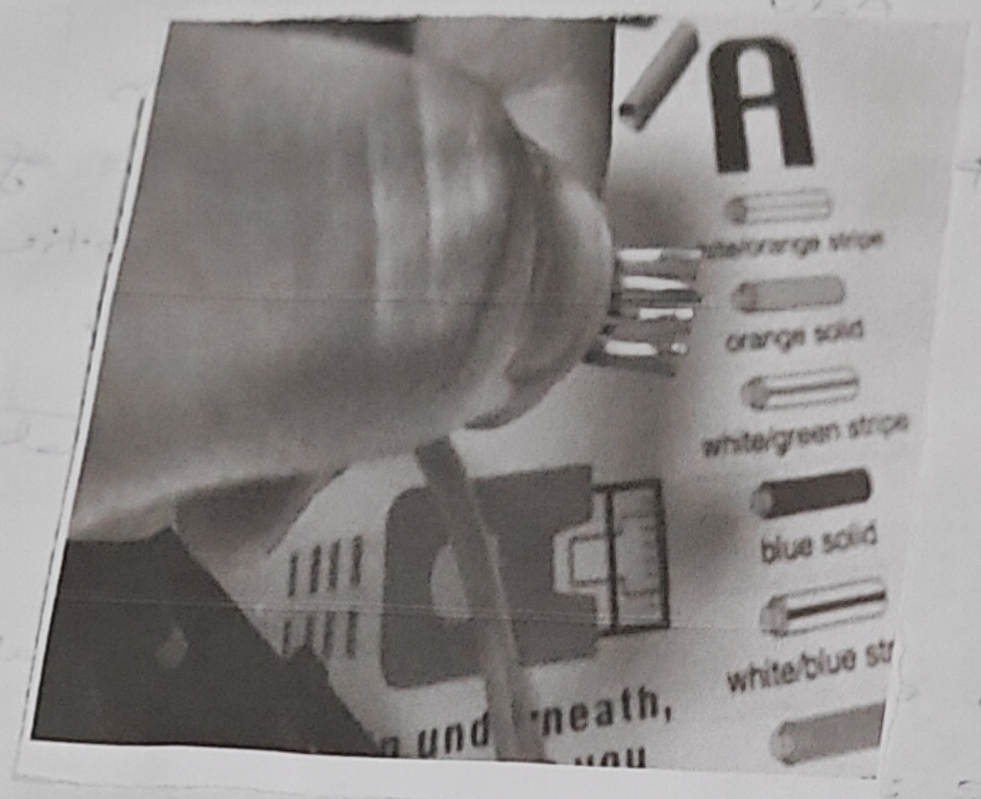


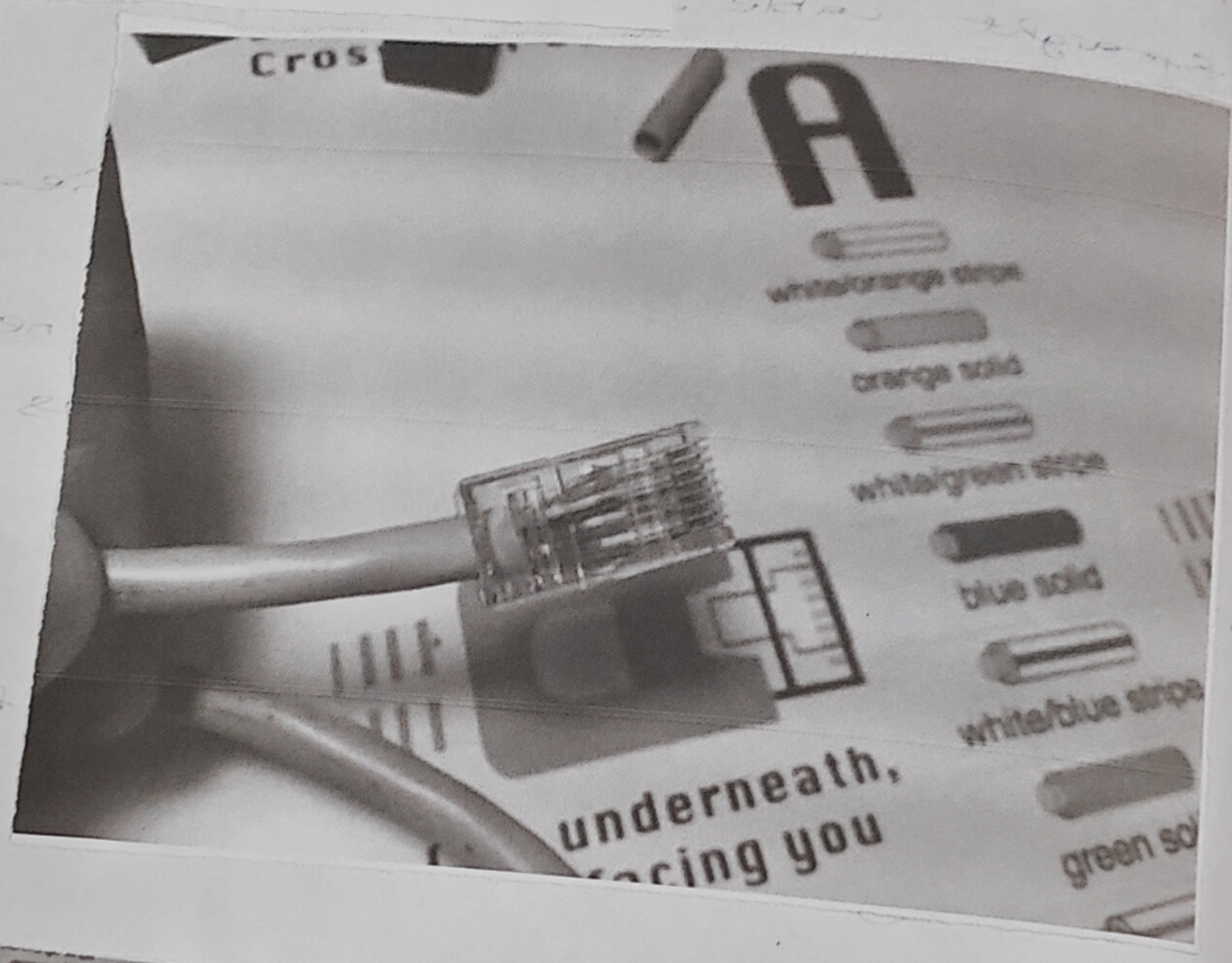


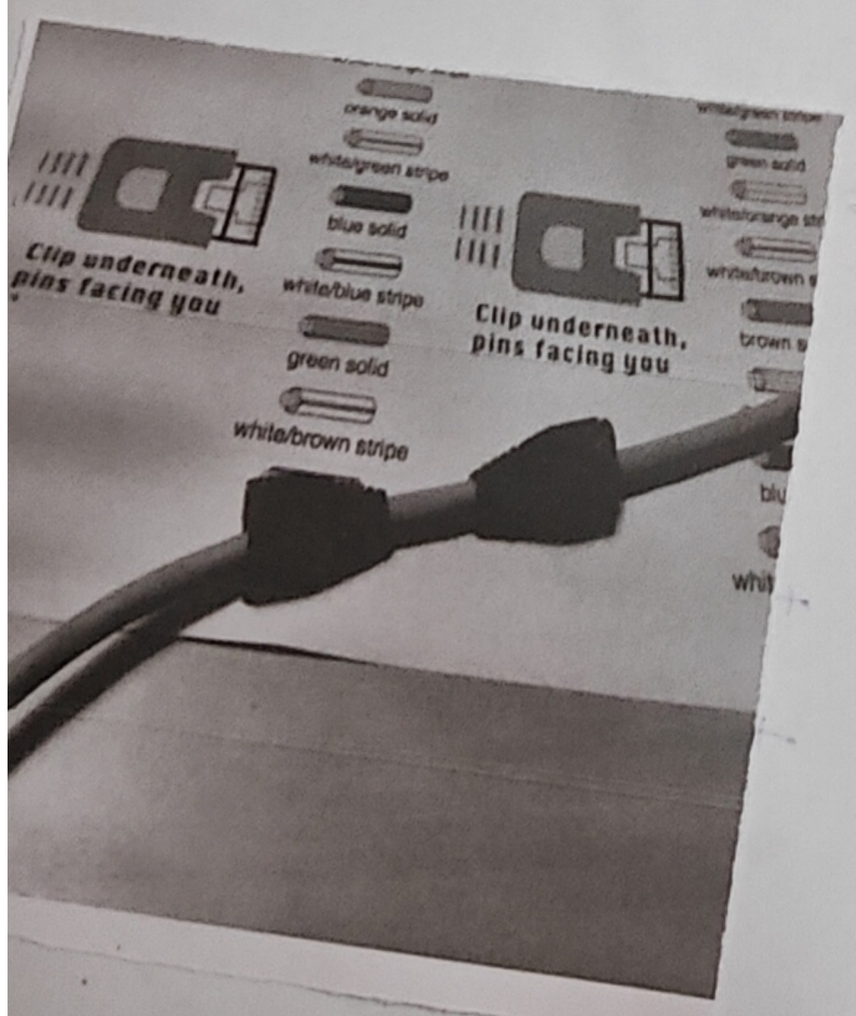
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Straight through network cable: both sides should be A
Crossover cable: One side A, one side B




A
orange solid
white/green stripe
blue solid
underneath,


A
blue solid
white/blue stripe
underneath,


orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Clip underneath, pins facing you

Step 4 :-

Once order is correct, Bunch them together in line and if there are any that stick out farther than others, snip them back to create even level.

Step 5 :-

Next, push cable right in. The notch at the end of plug needs to be just over the cable shielding

Step 6 :-

After the wires are securely sitting inside the plug, insert it into crimping tool.

Step 7 :-

Lastly, repeat for other end using diagram.

Result :-

Hence studied the different types of network cables.